Dynamic Response of an Offshore Jacket Structure

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PROBLEM STATEMENT, CRITERIA AND CONSTRAINTS

An offshore Jacket structure designed for a wind turbine in the Arabian Gulf.
It is a four legged structure made with stainless steel with a base to hold the wind turbine.
The wind turbine has a rating pf 5MW, it is supported by a pipe that is 68m long that is on a transition piece that is 4m height. The transition piece is on a 50m high jacket structure with a base of 30m x 40m and a top platform of 15m x 20m

PROPOSED SOLUTION & DESIGN

After all of the simulations are done on the proposed designs three designs are chosen based on the weight and First 3 modes of frequency
The three designs are; AB2 (X braces design) FS2 (V braces Design), and FS3 (Diamond Braces Design)

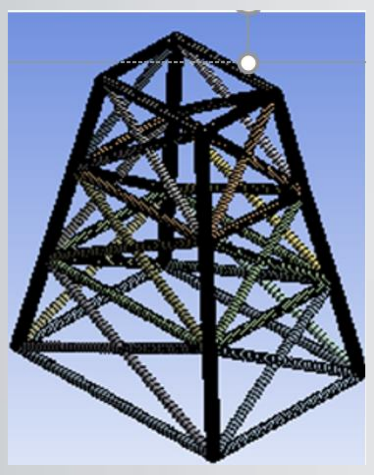
Figure1: AB2 Design (X braces

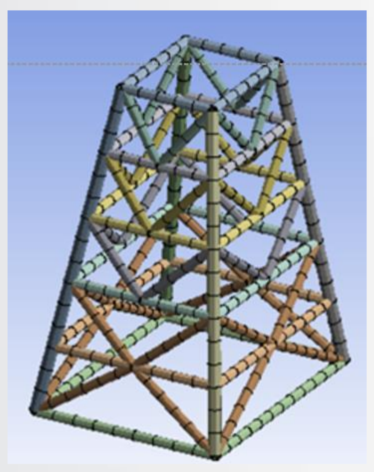
Figure2: FS2 design (V braces)

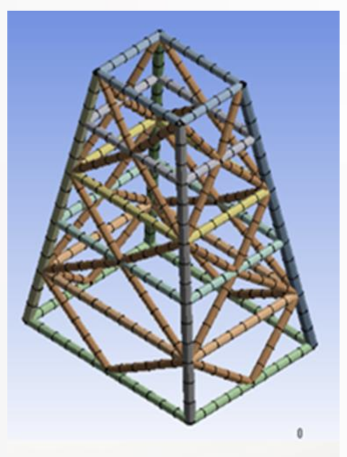
Figure3: FS3 design (Diamond Braces)

EXISTING SOLUTION AND LIMITATION

The designs went through 3 cases and a comparison was held between them.

Table 1: Case 1 of the cross section

Case 1	Ro	Ri
Braces	0.75 m	0.73m
Vertical lines	0.8m	0.77m
Legs	1m	0.95m

Table 2: Case 2 of the cross section

Case 2	Ro	Ri
Braces	0.78 m	0.76m
Vertical lines	0.83m	0.8m
Legs	1.03m	0.98m

Table 3: Case 3 of the cross section

Case 3	Ro	Ri
Braces	0.72 m	0.7m
Vertical lines	0.77m	0.74m
Legs	0.97m	0.92m

Table4: comparison between the designs with each case implied

Design	1st mode (Hz)	2nd mode(Hz)	3rd mode(Hz)	Weight (Tons)
AB2 (1)	4.8988	5.1904	5.496	1535.9
FS2(1)	4.6171	5.1538	5.1648	1704.2
FS3(1)	4.5457	5.1077	5.3145	1711.6
AB2 (2)	4.9531	5.382	5.522	1592.1
FS2 (2)	4.6487	5.1751	5.3476	1768.8
FS3(2)	4.5784	5.1309	5.5016	1776.4
AB2 (3)	4.8294	4.9977	5.3935	1479.7
FS2 (3)	4.5895	5.019	5.1354	1658.9
FS3 (3)	4.5084	5.083	5.1264	1650.7

IMPLEMENTATION AND RESULTS

Based on the previous Comparisons, the selected model is AB2 which is the X braces design, it was selected because it had the lightest weight and the highest frequency.

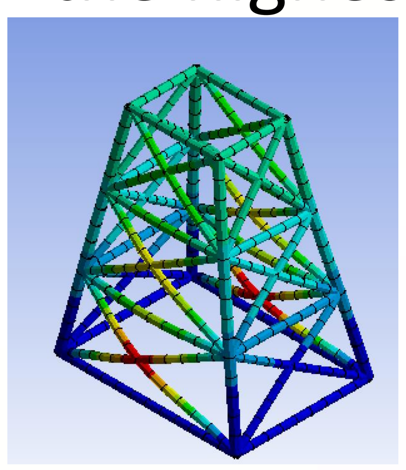
Figure4: AB2 first mode of deformation

	Mode	Frequency [Hz]
1	1.	4.8294
2	2.	4.9977
3	3.	5.3935
4	4.	5.4679
5	5.	5.8565
6	6.	5.8565
7	7.	5.8588
8	8.	5.8588
9	9.	6.2502
10	10.	7.0345

Figure5 : First 10 modes of Frequency, AB2 design

CONCLUSIONS AND FUTURE WORK

In Conclusion, based on the information we took from the Literature survey, we based our work with the methods they used. We had many designs , then based on the weight and frequency we choose three designs. The final design is AB2 and it is chosen because it has the lightest weight and the highest frequency.

REFERENCES:

1. Elshafey, A. A., Haddara, M., & Marzouk, H. (2009, January 1). Dynamic response of offshore jacket structures under random loads: Semantic scholar. undefined. Retrieved October 26, 2021, from https://www.semanticscholar.org/paper/Dynamic-response-ofoffshore-jacket-structures-Elshafey-Haddara/133ff12e1e99882e5cd5bcbca31d9dd8e55c4b03 .